# CFD Simulation of Flow Through an Orifice Plate

## 1 Introduction

Orifice plates are commonly used devices for measuring flow rates in pipes. They consist of a thin plate with a concentric hole in a straight pipe section. As fluid passes through the constriction, a vena contracta forms downstream, creating a pressure drop that can be used to calculate the flow rate.

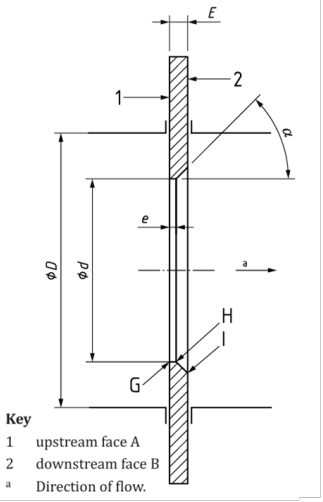
The objective of this work is to design an orifice plate for a 280 mm diameter ($D$) pipe, and to investigate the flow behaviour using computational fluid dynamics (CFD) software STAR-CCM+, with an interpretation of the results, while considering the assumptions and limitations of the approaches and models used. Experimental validation is planned at the TÜV high-pressure wet gas facility loop [1]. The experiment will use dry nitrogen at 61 bar (60 bar gauge pressure) with a flow rate of 1100 $m^3$/h at a temperature of 20°C.

## 2 Geometry Design

The design, installation, and use of orifice plate meters are governed by British Standards (BS). BS 5167-1:2022 covers the general theory and common requirements for pressure-differential flow meters, while BS 5167-2:2022 provides the requirements for plate geometry, upstream and downstream straight lengths, and the positioning of pressure tappings [2] [3]. Figure 1 shows the axial plane cross-section of a standard orifice plate, with key dimensions labelled.

According to Section 5.1.5 of BS 5167-2:2022, the plate thickness $E$ should not exceed 0.05$D$ and be larger than the orifice thickness $e$. Section 5.1.2.3 states that for pressures above 50 kPa, the plate must resist buckling. As the nitrogen is at 6.1 MPa, the upper thickness limit is chosen, $E$ = 14 mm. While this is a theoretical simulation, the design considers practical implementation. The orifice thickness $e$ must not exceed 0.02$D$, therefore $e$ = 5.6 mm is chosen with a bevel angle of $\alpha = 45°$, the midpoint of the permissible range given in Section 5.1.6.

Figure 1: Cross-section of a standard orifice plate




Section 5.1.8.1 states that the orifice diameter $d$ shall be no smaller than 12.5 mm, and the diameter ratio $\beta = d/D$ should lie between 0.10 and 0.75. $\beta$ = 0.6 is chosen, values less than 0.4 can create intense jets and large recirculation zones, resulting in steep gradients and increased difficulty for Reynolds-averaged Navier-Stokes (RANS) closure with higher sensitivity to mesh and near-wall treatment. $\beta$ = 0.6 gives an orifice diameter of $d$ = 168 mm.

Section 5.2 of BS 5167-2:2022 gives three pressure tapping options: corner tappings, flush with the orifice faces; flange tappings, 25.4 mm from the faces; and D and D/2 tappings, for this setup 280 mm upstream and 140 mm downstream of the orifice.

The D and D/2 arrangement is selected, as it measures pressures within the fully developed upstream flow and the recovered downstream flow. Flange tappings will likely be in zones of significant curvature in the pressure field, requiring a finer mesh to maintain accuracy. The tappings are assumed to be flush with the pipe wall and are therefore not modelled.

Table 3 of BS 5167-2:2022 states the required straight lengths between the orifice plates and upstream and downstream fittings, such as pipe bends. This simulation is of a straight pipe; therefore, no recommended values apply. Both the upstream and downstream lengths are initially set to an arbitrary length of 10$D$, with preliminary simulations used to verify whether this is a sufficient length for flow development.

# 3 Meshing Strategy

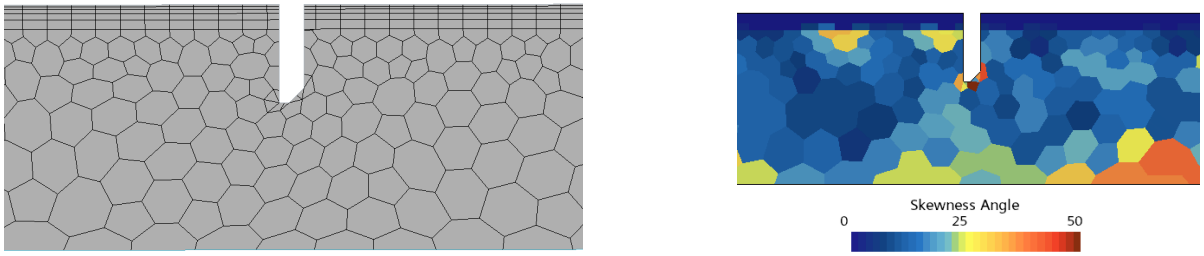
An axisymmetric 2D model was created in StarCCM+, as the geometry, boundary conditions, and mean flow field are rotationally symmetric about the pipe centreline. In turbulent flow, the Reynolds-averaged Navier-Stokes (RANS) approach represents time-averaged quantities, so the mean flow is axisymmetric. This approach has been used successfully in similar studies [4] and allows finer mesh resolution near the plate edge, wall boundary layers, and vena-contracta for the same overall cell count, improving near-wall accuracy.

A 2D automated mesh is set up using the Polygonal mesher and Prism Layer mesher. A Polygonal mesher was selected over a Quadrilateral mesher, as the latter only works well when the flow crosses the cell boundaries at right angles, which is not the case near the orifice or in recirculating regions. Polygonal cells also achieve comparable accuracy with fewer elements, reducing computational cost.

The Prism Layer mesher was applied near the pipe wall to resolve high velocity gradients near the pipe wall. These layers behave similarly to quadrilateral cells, as the velocity normal to the wall is negligible. Custom surface controls were used to exclude prism layers at the orifice and to suppress the default mesh refinement along the inlet, outlet, and symmetry boundaries.

The initial mesh was generated with a coarse resolution to enable quick simulation while providing sufficient information to guide refinement for further simulations. The base size was set to 0.14 m (equal to the pipe radius), with a target surface size of 10% and a minimum surface size of 1% of the base. The surface growth rate was 1.2, the upper recommended limit. The total prism layer thickness was set to be large at 10% of the base size (14 mm) with five layers in total.

The coarse mesh near the orifice is shown in Figure 2a, where it exhibits poor cell quality around the orifice edge. This mesh contained 4169 cells and took 0.78 seconds of computation time. As seen in Figure 2b, high levels of skewness occur at the orifice tip and along the axisymmetric place, with a maximum of 50°. The results of the first simulation, and further mesh refinements are discussed in Section 5.




(a) First coarse mesh, with poor quality mesh around the orifice.

(b) Skewness angle

Figure 2: Area of first coarse around the orifice of the axisymmetric pipe setup

# 4 Choice of Physics Model

A turbulent flow solver is used as the flow within the pipe is always turbulent. Pipe flow is considered turbulent when the Reynolds Number $R_e$ is greater than 4000. Using the mean flow 4.96m/s, and nitrogen properties at 61 bar and 20°C: density 69.486kg/m$^3$ and dynamic viscosity 18.674x10$^{-6}$ Pa s, the Reynolds number is $R_e = \frac{\rho U D}{\mu}$ = 5.17x10$^6$ [5].

The SST $k$-$\omega$ turbulent model was chosen over the $k$-$\epsilon$ model. The $k$-$\epsilon$ model performs well for external flows and free streams, but is less accurate in regions of strong pressure gradients and flow curvature, which are present in this study. The $k$-$\omega$ model performs better for internal flows and separated regions. The SST (Shear Stress Transport) model combines both models, reducing their individual limitations, making it suitable for orifice flow.

The boundary conditions were defined as a velocity inlet, a pressure outlet, an axis condition along the pipe centreline, and wall conditions applied to the pipe and orifice surfaces. A turbulent velocity inlet profile is calculated using the power law profile. The centreline velocity is calculated to be $U_{max} = 6.07$m/s, giving the velocity profile implemented with a user defined field function: $U(r) = 6.07(1 - \frac{y}{0.14})^{\frac{1}{7}}$ [6].

A limitation of this method is that it produces an infinite velocity gradient near the wall, making it invalid in the viscous sublayer. Therefore, the inlet length was determined based on the distance required for the wall shear stress $\tau_w$ to reach a constant value, ensuring a fully developed turbulent profile before the orifice.

The flow was assumed to be steady-state, as the boundary and initial conditions remain constant. This can be validated if the simulation residuals decrease and level off without oscillation, which would be the indication of transient behaviour. Additionally, if the axial pressure gradient is equal to zero at the outlet with no backflow, the steady state can be assumed valid.

The flow can be assumed incompressible. For gases, this is valid when the mean velocity is less than 0.3 times the local speed of sound. The speed of sound of nitrogen at 61 bar and 20°C is 361.96 m/s. The threshold for incompressibility is therefore 108.59 m/s, a factor of 22 higher than the mean flow velocity. Flow velocity will be higher at the vena-contracta, but still far below the Mach number of 0.3.

In summary, the following physics models and assumptions were used:

1. Flow type: Steady, incompressible gas
2. Turbulence model: SST $k$–$\omega$
3. Material: Nitrogen ($\rho = 69.486$ kg/m³, $\mu = 18.674 \times 10^{-6}$ Pa s at 61 bar)
4. Reference pressure: 61 bar (irrelevant to solution since density is constant)

## 5 Simulations and Results

The first simulation took approximately 10 seconds to perform 500 iterations and produced the pressure and velocity vector plots shown in Figure 3 and 4. The simulation converged after 200 iterations, with 0.03 the highest residual value, 0.0006 the lowest; both above the acceptable convergence criteria.

Figure 3: Pressure plot of first simulation (relative)

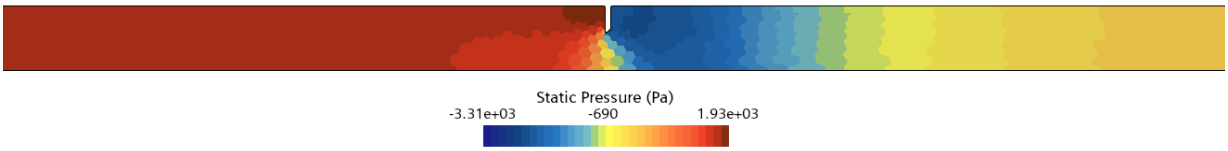
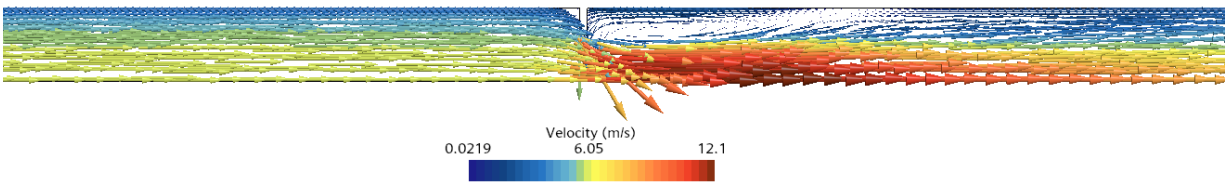



Figure 4: Vector velocity plot of first simulation

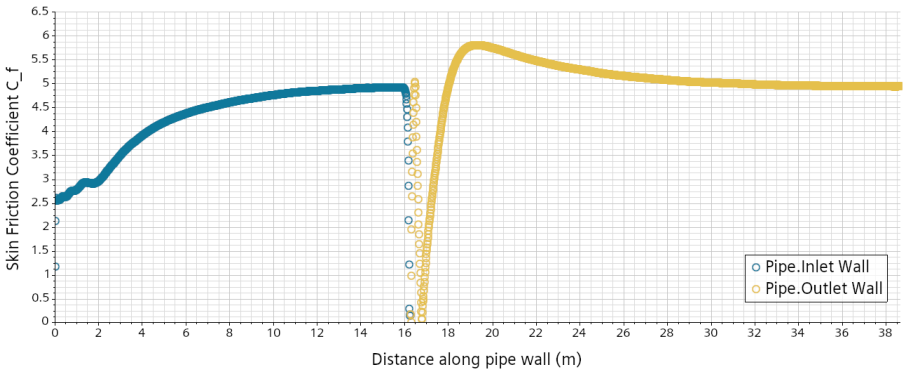



The results show a high pressure gradient along the $x$ direction that gradually recovers downstream, shown in Figure 3. Figure 4 shows a small recirculation area upstream of the orifice front face, and a much larger area of recirculation downstream. These features suggest the need for local mesh refinement to improve accuracy in these high-gradient regions.

To determine the required inlet and outlet lengths, the skin friction $C_f$ on the pipe wall was plotted as a function of $x$. Where $C_f$ plateaus, this is where the flow is fully developed and unaffected by boundary conditions. Simulations were repeated with progressively longer pipe lengths until the flow became

fully developed upstream and pressure recovery unaffected downstream. An additional distance of $5D$ is included before the pressure-outlet boundary to prevent any artificial back-pressure effects on the reattachment and redevelopment of the flow. The correct inlet length was found to be $58D$ (16.24 metres) and the outlet $80D$ (22.4 metres). A plot of $C_f$ along the pipe wall is shown in Figure 5.

Figure 5: Skin friction $C_f$ along the inlet (blue) and outlet (yellow)




To improve convergence, the prism layers were refined. The first layer was positioned at $y^+ \approx 1$ and the final layer within the logarithmic region at $30 < y^+ < 300$. The initial simulation gives wall shear stress as $\tau_w = 2.6$ Pa, giving $y^+ = 1 = 0.00137$ mm and $y^+ = 200 = 0.274$ mm. Therefore the total prism layer thickness was reduced from 5% of the base to 0.2%, and number of layers set to 12, for a growth rate of 1.5.

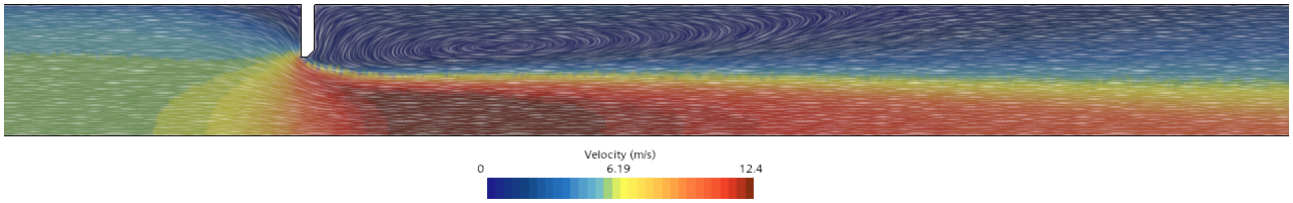
Based on these observations, a block volumetric control was applied across the region of high velocity and pressure gradients, including the areas of recirculation, as shown in Figure 6. The refinement used a target size of 5% of the base from $2D$ upstream to $8D$ downstream of the orifice. Additional refinement was tested in these regions, but further reduction of cell size caused residual oscillations, suggesting transient behaviour, potentially due to vortex shedding behind the orifice. Although the residuals converged when refined to 2% of the base size, they remained high. Increasing the prism layer thickness reduced oscillations but worsened convergence, particularly with the turbulent kinetic energy ($k$) residual. Therefore, the final configuration used a base size of 0.14 m with refinement set to 5% of the base size.

Figure 6: Mesh refinement around areas of recirculation, and high physical gradients



The refined grid contained 65,242 cells and took 6.92 seconds to mesh. The simulation took 4 minutes 54 seconds to complete 1500 iterations. Figure 7 shows the velocity field visualised using line integral convolution. A small area of recirculation forms between the pipe wall and the front face of the orifice, and a large recirculation area appears downstream before flow reattachment. Simulations using finer refinement (2% of the base), but with higher residuals, revealed a secondary area of recirculation behind the orifice.

Figure 7: Line integral convolution velocity plot of the refined mesh

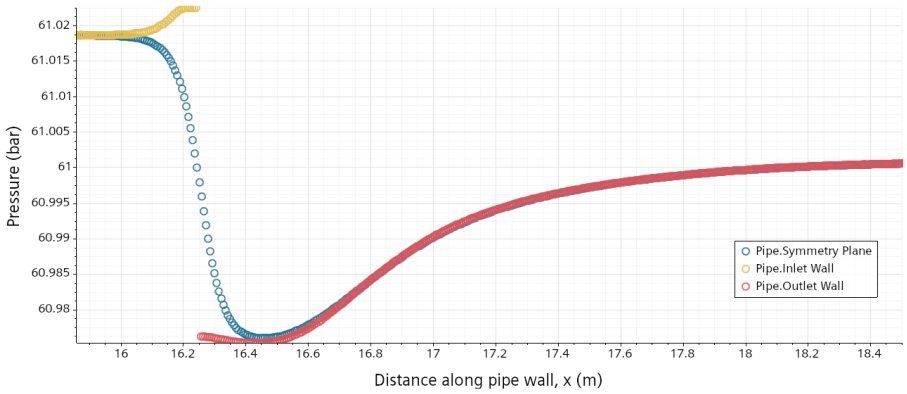



The flow of a turbulent pipe through an orifice plate is a well-documented case, with analytical models available for predicting the theoretical pressure differential using the $D$ and $D/2$ tapping arrangement. These equations are provided in BS 5167-2 2022. Equation (1) from the standard is used, rearranged for pressure differential, $\Delta p$.

First, calculating the discharge coefficient $C$ using the Reader–Harris/Gallagher correlation given by BS 5167-2 Eq.(4), gives a value of $C = 0.6055$. The expansibility factor $\epsilon$ is given by BS 6167-2 Eq. (5), but it requires the pressure ratio $p_2/p_1$, which is unknown. Using an iterative process using Microsoft Excel, we find $\epsilon = 0.99932$ and a pressure ratio of 0.99743. The ratio of specific heats used was $\kappa = 1.51$. This gives a pressure differential of $\Delta p =$ 15.656 kPa and the D/2 pressure tapping value to be $p_2 =$ 60.843bar.

From the CFD results (Figure 8), the relative pressure values at the tapping locations, $x$ = 15.96 m and $x$ = 16.394 m, are 61.0185 bar and 60.9752 bar, giving a pressure difference of 4.33 kPa, a 72.3% underprediction, when expected deviations are between 2% and 8% [7]. This suggests a critical flaw in either the theoretical calculation or the setup of the mesh, prism layers, or boundary conditions, and the simulation can not be used for design purposes. However, the general shape of the pressure and velocity plots agree with published academic papers, which indicates that the fluid physics are correct [7] [8].

Figure 8: Pressure along pipe wall and centreline across orifice




## 6 References


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